Detailed Project Report (DPR)

Agri-Biomass Biochar & Enriched Bio-Fertilizer — Green-Category Unit (starting 0.5 t/day, ramping to 2 t/day)

Promoter: Arun T. (sole proprietary concern) **Location:** Eruthenpathy, Chittur Taluk, Palakkad District, Kerala (Kerala–Tamil Nadu border) **Feedstock origin:** Own coconut-farm residues + open-market agri/wood biomass procured from multiple local sources **Structure:** A **single registered unit** that **starts small (~0.5 t/day) and ramps in place to ~2 t/day**, housed in a **purpose-built ~10,000 sq ft single clear-span hall (with 3 movable partition rails)** on the promoter's own 38-cent corner plot, with all bulk feedstock and drying handled on the promoter's **own 10-acre land ~200 m away**. Designed for the **Green** pollution category. **Prepared for:** Submission to District Industries Centre (DIC), Palakkad and lending bank (AIF / term loan) **Document status:** Planning-grade DPR — figures are indicative estimates to be firmed up with (a) a feedstock proximate-analysis lab report, (b) formal machinery quotations, and (c) a written KSPCB categorisation confirmation, before sanction.

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1. Executive summary

This project establishes an **agri-biomass biochar and enriched (microbe-inoculated) organic bio-fertilizer** business as a **sole proprietary concern of Arun T.**, as a **single unit that starts small and grows in place** at Eruthenpathy, Chittur, Palakkad.

- **Constitution:** sole proprietorship — **Arun T., proprietor**.
- **One unit, phased in place (not split):** start at **~0.5 t/day biochar (500 kg/day)** and **ramp to ~1 t/day and then ~2 t/day** by adding modular reactor capacity within the **same registered premises**.
- **Purpose-built hall:** a **~10,000 sq ft single clear-span covered hall** (over ~23 cents of the 38-cent plot) is built **once, up front, for the 2 t/day endgame**, fitted with **3 movable partition rails** (sliding partitions on floor tracks) that **open or close to reconfigure the floor into larger or smaller working bays** — process, curing, packing and storage — as throughput ramps. Building the full civil once (rather than in stages) also lets the project **capture the maximum Kerala ESS capital subsidy**.
- **Land:** the **own 38-cent corner plot** carries the hall and process; the promoter's **own 10-acre land ~200 m away** handles **all feedstock aggregation, sun-drying, chipping and bulk storage**.
- **Feedstock: multi-source, own-farm + open-market agri/wood biomass** — coconut fronds, petioles, husk, shells and spathe; coir pith and shells from local dehuskers/coir units; and open-market residues (sawdust/wood chips, cashew shell, areca husk, paddy husk, groundnut shell, prunings).
- **Pollution category — Green (target).** Designed as *general biomass biochar (soil-amendment) production*, which the KSPCB revised categorisation list (as on 29.12.2025) places in the **Green** category — distinct from "charcoal from coconut shell by pyrolysis" (Orange). See Section 7.
- **Institutional tailwind:** the project plugs into the **Kerala Government biochar expert committee** (constituted on a proposal of the **Australia India Business Council, AIBC**) and takes technical help from **CPCRI** and **KAU**. See Section 16.
- **Project cost (Phase-1):** **~₹1.09 crore** (~₹1.01 cr fixed capital + ₹8 L working-capital margin); land is the promoter's own (in-kind, excluded).
- **Financing:** promoter equity + **AIF term loan (3% interest subvention, 7 yrs)** + **CGTMSE** collateral-free cover + **Kerala ESS capital subsidy (~₹40 L — the scheme maximum)**.
- **Markets: domestic (India) + international**, plus **carbon-removal credits** (the highest-value stream) via an aggregator (Puro.earth CORCs). Australia/NZ is one explorable export option (Section 16).
- **Indicative payback:** the front-loaded hall lengthens the Phase-1 physical-only payback, but returns strengthen sharply as the unit ramps to 2 t/day and once carbon-credit revenue is realised.

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2. Promoter

Promoter / constitution: Arun T. — sole proprietary concern (proprietorship).

- Owns the project land: a **38-cent corner plot** (the manufacturing hall) at Eruthenpathy, and **~10 acres ~200 m away** used for feedstock, drying and storage. Together they comfortably meet all of the project's land needs.
- Contact: 4/296, Mangapallam South, RVP Pudur Post, Palakkad District, Kerala – 678555 · 9901955667 · er.arunt@gmail.com.
- Captive demand: the promoter's own coconut holding consumes and demonstrates the enriched product, de-risking market entry.

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3. Project concept & rationale

Agri-biomass residues (coconut fronds/husk/shell, coir pith, wood/sawdust and other crop residues) are abundant, low-value and often burnt in the open. Converting them to biochar via controlled, low-emission pyrolysis:

1. Diverts local biomass waste from open burning into a productive use.
2. Produces a stable, carbon-rich **soil amendment**; enriched with organics + microbes it becomes a branded slow-release **bio-fertilizer** (a soil input — **not** fuel charcoal and **not** activated carbon).
3. Generates durable **carbon-removal credits** (1 t biochar ≈ ~2.2 t CO_2 sequestered for centuries).
4. Qualifies for strong central + state subsidy support as an agro-based / bio-fertilizer MSME.
5. Is structured to be **Green-category**, so the unit can be established under a light-touch consent regime and grown in place (Section 7).

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4. Market analysis & best markets

Biochar has three revenue layers — physical char, enriched product, and the carbon-removal credit — sold across domestic and international markets. Marketing is primarily **domestic + international**; the carbon credit is the highest-value, inherently global stream.

Indicative prices (2026 references):

Segment	Price	Note
Bulk agricultural biochar	US$400–700 /t (₹35+ /kg blended, India)	Coarse, basic feedstock
Premium horticultural biochar	US$800–1,500 /t	Sieved, low-ash, inoculated
Activated / specialty	US$1,500–3,000 /t	High surface area
Biochar CDR carbon credit	**US$100–250 /t CO_2** (~US$150 avg)	Separate from physical char; Puro/EBC command 15–25% premium

Best markets, best-first:

1. **Carbon-removal credits (global — highest value).** ~US$150/t CO_2; **supply is severely constrained** (≈89% of 2025 supply pre-committed), buyers concentrated among **Microsoft, Google, Frontier, JPMorgan, Swiss Re, Stripe**. The **EU CRCF** now recognises biochar as permanent removal. Routed via an aggregator (Section 15) under **Puro.earth/EBC** — a premium, seller-favourable stream from day one.
2. **Domestic India (primary physical volume).** Agriculture is ~73% of biochar demand. Nearest, lowest-logistics market: **acidic laterite soils (Kerala), plantation & horticulture crops, nurseries, potting mixes, and the organic-input market**; anchored by own-holding demonstration.
3. **Premium physical export — Europe & North America. Europe** (EBC-certified, EU CRCF, Germany hub) commands the **highest physical prices**; **North America** is the **largest** biochar market. **EBC/IBI certification** unlocks both.
4. **Australia / New Zealand (explorable option).** Feed-additive + soil-amendment demand, pursued through the **AIBC / Kerala expert-committee** channel (Section 16); needs **ANZBIG Code of Practice** alignment + Australian **BICON** import clearance.

Demand strategy: anchor on (a) own-holding consumption and domestic enriched/branded sales, (b) the **carbon credit** via an aggregator (premium, global), (c) **EBC-certified premium export** to Europe/North America, and (d) Australia/NZ as an explorable add-on. Sell *enriched/certified* product (margin), not raw char (commodity).

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5. Feedstock plan (multi-source)

The unit runs on a mix of own-holding and open-market biomass — giving supply security and reinforcing the *general biomass* (Green) classification.

Source	Material	Basis
Own coconut holding	Fronds, petioles, husk, shells, spathe	Free / captive
Local dehuskers & coir units	Coir pith, coconut shells	Open-market purchase
Local sawmills / carpentry	Sawdust, wood chips, offcuts	Open-market purchase
Local farms / traders	Cashew shell, areca husk, paddy husk, groundnut shell, prunings, seasonal residue	Open-market purchase / collection

All aggregation, sun-drying, chipping/shredding and bulk storage are done on the **own 10-acre land (~200 m)**; only a 1–2 day day-bin sits at the hall. Feedstock is documented **residual/waste biomass** for carbon additionality. Execute **supply MoUs with 3–4 local sources** plus own-holding residue.

6. Technical design & capacity (single unit, phased in place)

6.1 Ramp path (one unit, capacity added in place — not split)

Phase	Biochar	Feed (dry biomass)	Annual biochar	CO_2 removal
Phase-1 (start)	**~0.5 t/day (500 kg/day)**	~2 t/day	~150 t/yr	~330 t CO_2/yr
Phase-2	~1.0 t/day	~4 t/day	~300 t/yr	~660 t CO_2/yr
Phase-3 (endgame)	~2.0 t/day	~8 t/day	~600 t/yr	~1,320 t CO_2/yr

Capacity is grown by **adding modular reactor capacity inside the same hall** and up-rating the shared gas train — the unit is never split into separate registrations. The **hall, utilities and emission stack are built once for the 2 t/day endgame**, so ramping needs only the extra reactor module(s).

6.2 Emission control & Green-category design

Closed, **continuous/semi-continuous** pyrolysis with a **syngas afterburner/flare** (self-fuelling the process and the dryer) → **cyclone/dust collector** → **alkaline scrubber with mist eliminator** → **stack** meeting KSPCB norms; **real-time temperature monitoring** and emergency shutdown; **Zero-Liquid-Discharge** (tar condensate and scrubber blowdown captured and reused — no effluent to land or water). **No open burning** at any stage. The design is built to the **KSPCB "Environmental Guidelines for Charcoal Manufacturing Units" dated 02.12.2025**.

7. Pollution-control category & consent strategy

Objective: establish the unit in the Green category — the lenient regime — legitimately, on the merits of the process.

7.1 Why Green applies

The KSPCB revised categorisation list **as on 29.12.2025** classifies *general charcoal / char (biochar) production* under **Green**, distinct from the *"Charcoal Production from Coconut Shell by Pyrolysis"* **Orange** entry (Notification No. KSPCB/159/2022-SEE-3 dated 10.08.2024). Our unit fits the **general / Green** description because:

- **Feedstock is mixed agri-biomass**, not a coconut-shell-only charge.
- **Product is biochar for soil**, not fuel charcoal and not activated carbon.
- **No pyrolysis-oil or carbon-black recovery** and **no chemical/steam activation**.
- **Low pollution potential** — closed process with afterburner, scrubber, controlled stack emissions and **Zero-Liquid-Discharge**.
- **Guideline-compliant** with the 02.12.2025 charcoal-manufacturing guidelines.

7.2 Consent path

- **Green-category CTE then CTO** — no public hearing.
- **No Environmental Clearance** (biochar from agri-biomass is not EIA-2006 scheduled; not Red).
- Only the **hall plot** (pyrolysis) needs consent; the off-site drying/storage yards on the 10-acre land are **non-polluting ancillary** areas.
- A **written KSPCB clarification** confirming the Green entry for this unit has been sought (separate category-clarification letter); sanction should follow that confirmation.

8. Machinery & suppliers (Phase-1 ~0.5 t/day, expandable)

Local (Coimbatore–Pollachi hub, 30–60 km — preferred for service/spares): coconut-shell/biomass carbonization plant manufacturers (e.g., Micro Fab Engineers, Essar Engineers, Veera Biopower and similar). Indicative small semi-auto units 0.5–1 t/day-class ≈ ₹6–25 L.

National (continuous, low-emission, certification-grade — better for EBC/carbon): - **Kerone Engineering (Mumbai)** — continuous multi-feedstock pyrolysis, waste-heat drying, cyclone + scrubber, SCADA. - **W2E Innovations (Ahmedabad/Kharagpur; Bengaluru branch)** — integrated biomass→biochar + wood-vinegar + fertilizer, "verification-ready." - **Susstains Engineering (Karnataka)** — IIT-Madras-derived zero-smoke continuous process (~34% yield). - **Ylem Energy (New Delhi)** — smaller batch furnaces (lower cost; batch → less consistent).

Recommendation: a **compact continuous/semi-continuous ~0.5 t/day plant with integrated dryer + afterburner + scrubber**, with **pre-provisioned tie-ins for Phase-2/3 reactor modules** so the unit grows in place inside the hall. Budget ~₹22 L for reactor + gas train + dryer at Phase-1 (see Annexure D).

9. Land, layout & civil works

Two own plots meet all needs:

Plot	Role	Activities
Own 38-cent corner plot (roads on south & east)	**Manufacturing hall (registered; needs CTE/CTO)**	~10,000 sq ft single clear-span hall with 3 movable partition rails (reconfigurable bays: process, curing, packing, storage); emission stack; office/lab/weighbridge
Own ~10-acre land (~200 m)	Non-core	Feedstock aggregation, **sun-drying**, chipping/shredding, bulk biomass storage

The hall (~10,000 sq ft over ~23 cents): a **single clear-span covered hall** built for the 2 t/day endgame. **3 movable partition rails** (sliding partitions running on floor tracks) divide the clear span into **reconfigurable working bays** — the operator can **open the rails to run the whole floor as one large bay, or close them to separate process, curing, packing and storage bays** — resized as production ramps from 0.5 to 2 t/day. Office/lab, weighbridge and utilities sit at the frontage. Building the full civil once avoids re-construction on ramp and captures maximum ESS. Balance of the plot (~15 cents) is greenbelt, internal roads, parking and setbacks; the corner location (roads on **south and east**) allows separate material-in and dispatch gates.

Pre-checks: land-use where required; KSPCB setback from residences/water bodies/eco-sensitive zones; road access from the south/east frontage; 3-phase power; water.

Construction (hall + site works, ~10,000 sq ft): ≈ **₹54 L** hall (clear-span + 3 movable partition rails) + ~₹6 L site development (see cost table).

10. Regulatory & statutory compliance

Approval	Authority	Notes
Udyam (MSME) registration	Online	Gateway for schemes
Consent to Establish (CTE)	KSPCB	**Green category**
Consent to Operate (CTO)	KSPCB	Green; amend upward on ramp
Panchayat D&O (Dangerous & Offensive) licence	Grama Panchayat	Manufacturing hall
Building permit	Grama Panchayat	Hall/plant
Fire NOC	Fire & Rescue Dept.	Thermal process; large covered hall
GST registration		
Land-use conversion	Revenue Dept.	If plot is agricultural

Operating without CTE/CTO is a criminal offence under EPA 1986. No open burning of biomass at any stage. Environmental Clearance is not required (non-EIA-scheduled, non-Red).

11. Cost of project (Phase-1)

#	Item	₹ Lakh
1	Covered hall ~10,000 sq ft — single clear-span + 3 movable partition rails (built for the 2 t/day endgame)	54
2	Pyrolysis reactor ~0.5 t/day (modular, expandable to 2 t/day)	16
3	Emission control — afterburner/flare + cyclone + scrubber + stack (Green-compliant)	6
4	Char cooling, sieving, enrichment mixer, bagging	5
5	Electrical, water, fire, weighing	5
6	Feedstock handling on 10-acre land: sun-drying floor, chipper/shredder, storage	5
7	Lab / QC + office	2
8	Site development (corner plot: compound wall, gates on S & E, paving, greenbelt)	6
9	Preliminary & pre-operative (CTE/CTO, DPR, EBC/MRV)	2
	Fixed capital sub-total	**101**
10	Margin money for working capital	8
	Total project cost (Phase-1)	**~₹1.09 crore**

Land is the promoter's own (in-kind) and is **excluded**. **Ramp add-ons (from cash flows / top-up AIF loan):** Phase-2 (to 1 t/day) ≈ **+₹22 L**; Phase-3 (to 2 t/day) ≈ **+₹40 L** (extra reactor modules only — the hall, stack and utilities are already built).

12. Means of finance & subsidy stack (Phase-1)

Source	₹ Lakh	%
Promoter equity (cash)	27	25%
Term loan — bank, under **AIF (3% interest subvention, 7 yrs)**, **CGTMSE** collateral-free	74	68%
Working capital limit (bank cash credit)	8	7%
Total	**109**	100%

Kerala ESS capital subsidy — ~₹40 L (the scheme maximum). At the ~45% band, the cap of ₹40 L is reached at ~₹89 L eligible fixed capital; the project's ~₹101 L fixed capital **comfortably maxes the subsidy**. ESS is received as **reimbursement after commissioning** and applied to **prepay the term loan**, cutting effective debt to ~₹34 L.

Subsidy stack logic

- **Anchor = Kerala ESS** (state capital subsidy). Target **45%** band: 25% base (young/women/SC-ST/NRK) + 10% priority sector (agro-based / bio-fertilizer) + 10% new technology. *Palakkad does not get the extra 10% district bonus.* The **large hall lifts fixed capital past the ₹89 L point, so the full ₹40 L cap is captured.**
- **+ AIF** (central): 3% interest subvention + CGTMSE guarantee — different basis, **stacks legitimately with ESS**; confirm convergence in writing with DIC + bank.
- **+ CGTMSE:** collateral-free loan → the land is not mortgaged.
- **Do NOT use PMEGP** — it disqualifies units that have availed other government subsidy, voiding ESS + AIF.
- **Effective term-loan interest:** 9% − 3% = **~6%**.

13. Working capital (indicative, annual, Phase-1)

Feedstock collection/purchase/transport, enrichment inputs (poultry manure, neem cake, rock phosphate, microbial consortium), packaging, wages, power, and ~1–2 months finished-goods holding. Phase-1 operating cost ≈ **₹30 L/yr**; scales with capacity on ramp.

14. Profitability & ROI (Phase-1 and endgame)

Common assumptions: 300 operating days; biochar 1 t ≈ 2.2 t CO_2; carbon net realization ~₹10,000/t CO_2 (after aggregator/MRV); enriched blended price ₹35/kg. Effective interest 6%.

Scenario	Biochar	Revenue	Op. cost	Net profit (approx)	Payback
Phase-1 — base (physical only)	150 t	₹52.5 L	₹30 L	**~₹12 L**	~5–6 yr*
Phase-1 — + carbon	150 t	₹52.5 L + ₹33 L = ₹85.5 L	₹32 L	**~₹40 L**	~2 yr
Endgame 2 t/day — base (physical only)	600 t	₹2.10 cr	₹1.05 cr	**~₹80 L**	—
Endgame 2 t/day — + carbon	600 t	₹2.4 cr + ₹1.32 cr = ₹3.72 cr	₹1.20 cr	**~₹2.1 cr**	<1 yr

*The Phase-1 physical-only payback is longer because the full-size hall and utilities are built up front for the 2 t/day endgame and to capture maximum ESS. Returns strengthen sharply as capacity ramps inside the same hall (each added reactor module needs little fresh civil) and once carbon revenue is realised. Net capital at risk (Phase-1) ≈ ₹70 L after ESS ₹40 L; DSCR (Year 2) ≈ 3.3** (bankable). Treat physical-only as the base case; carbon is upside pending a signed aggregator offtake.

15. Carbon-credit strategy & aggregator options

At ~330 t CO_2/yr (Phase-1) rising to ~1,320 t CO_2/yr (endgame), volumes are **too small to transact directly with corporate buyers** initially (~85% of CORCs are sold via brokers/aggregators). Route through an **aggregator/developer** who provides MRV, certification (Puro.earth CORC methodology) and access to the concentrated global buyers (Microsoft, Frontier, etc.).

Party	Role / fit
Varaha (Gurugram/New Delhi) — India's largest biochar CDR developer; Puro.earth + Verra; offtakes with Google, Microsoft, Kellanova	**Best first contact.** Digital MRV + credit commercialization; issues CORCs.
Equilibrium — industrial biochar operator	Developer/partner; potential offtake channel.

Party	Role / fit
Eco Saarthi — cooperative aggregating agri residue → biochar	Fit for an aggregation structure.
Puro.earth	The **registry/standard** (CORCCHAR methodology); certification via a developer.
BioFlux (advisory)	Independent biochar-CDR advisory — feasibility, MRV & certification, buyer alignment.

Action: approach **Varaha first** for MRV + offtake; target a **Letter of Intent (LOI)** as volumes ramp. Document feedstock as **residual/waste biomass** for additionality.

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16. Institutional collaboration — AIBC, the Kerala biochar expert committee, CPCRI/KAU

Kerala's biochar push has a **direct channel the project should leverage**:

- **How it began:** the **Australia India Business Council (AIBC)** — which promotes Australia–India bilateral trade — proposed the initiative (letter dated **15 July**), leading the **Kerala Government to constitute a 7-member biochar expert committee**, **chaired by the Special Officer for Agriculture (WTO Cell)**, with experts from **Kerala Agricultural University (KAU)** and the **Central Plantation Crops Research Institute (CPCRI)** and Agriculture Department officials. Its brief: study **biochar and derivatives from coconut biomass** for domestic use and export.

How this project takes help:

1. **Plug into the State expert committee (WTO Cell).** Write to the Special Officer for Agriculture (WTO Cell) offering this unit as an **on-ground pilot / demonstration** feeding the committee's study — for State recognition, policy alignment (including the **Green-category** treatment), and credibility with buyers and lenders.
2. **Technical help from CPCRI (Kasaragod) and KAU (Vellayani).** CPCRI has pyrolysed **coconut husk, petiole and coir-pith into biochar** (characterisation, soil-amendment validation); KAU has field trials of **coconut-frond biochar + liming on Kerala laterite soils**. Seek support for **feedstock-specific recipes, product characterisation/QC, agronomic validation and farmer demonstrations** — de-risking quality and speeding EBC/organic certification.
3. **Export facilitation via the AIBC.** Use the AIBC channel to explore **Australia/NZ** demand (feed-additive + soil amendment) — one option alongside the higher-value carbon-credit and Europe/North-America premium routes (Section 4).
4. **Standards & biosecurity for any Australia/NZ export.** Align to the **ANZBIG Code of Practice** + EBC; clear **Australian biosecurity import (BICON)** — carbonised biochar from plant material needs an **import permit** and biosecurity compliance.

Actions: (a) letter to the Special Officer (WTO Cell) to enrol as a pilot; (b) technical-support/MoU requests to CPCRI and KAU; (c) engage the AIBC to scope Australia/NZ market access.

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17. Value addition — enriched bio-fertilizer

Base biochar is K-rich; enrich to a balanced slow-release organic input: - **Nitrogen/phosphorus:** poultry manure, neem cake, bone meal / rock phosphate. - **Biology (branded premium):** microbial consortium — Azospirillum/Azotobacter (N-fix), PSB/phosphobacteria, potash mobilizers, Trichoderma, mycorrhiza (VAM). - **Base/biology co-compost:** vermicompost + cow dung; Kerala liquid organics (fish amino acid, panchagavya, jeevamrutham); seaweed extract for micronutrients. - **Methods:** nutrient impregnation/soaking then low-temp drying; or co-composting/vermicomposting with ~10% biochar (in the curing/packing hall); pelletize to cut leaching. - **Certifications for premium/export:** EBC, IBI, ANZBIG Code of Practice (for Australia), organic-input (Jaivik Bharat / NPOP for export).

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18. Implementation schedule (sequence, not calendar)

1. Feedstock lab proximate analysis (moisture, fixed carbon, ash, K) — decides final Phase-1 sizing.
2. Udyam registration; DIC/ESS pre-consultation; feedstock MoUs (3–4 local sources).
3. **Written KSPCB categorisation confirmation** (Green); enrol with the **WTO-Cell expert committee** as a pilot.
4. **CTE (Green)**; hall design/build (10,000 sq ft single clear-span, 3 movable partition rails, endgame-sized); machinery order (~0.5 t/day reactor + gas train + dryer, with ramp tie-ins).
5. Hall + site works on the corner plot; feedstock yards on the 10-acre land.
6. Panchayat D&O + building permit + fire NOC; **CTO**.
7. AIF term loan + CGTMSE; commissioning; **ESS claim (max ₹40 L)**.
8. Product validation on own holding; **CPCRI/KAU** characterisation; EBC/organic certification.
9. Engage carbon aggregator (Varaha) and, for exports, EBC certification + the AIBC channel.
10. **Ramp in place** to 1 t/day then 2 t/day from cash flows / top-up loan.

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19. Risks & mitigation

Risk	Mitigation
Pollution category	Design for Green on the merits; **written KSPCB confirmation**
Front-loaded civil (big hall)	Built once for the 2 t/day endgame + maxes ESS; ramp needs little fresh civil; carbon revenue upside
Feedstock moisture / drying energy	Sun-dry on the 10-acre land; waste-heat dryer; lab-test first
Feedstock security	Multi-source (own holding + 3–4 open-market suppliers)
Demand-side	Domestic anchor + carbon credit (global) + EBC premium export (Europe/N. America); Australia/NZ explorable
Export/biosecurity	EBC/ANZBIG certification; early **BICON** enquiry for Australia
Carbon-credit dependency	Bank case = physical only; carbon = upside pending LOI
Product consistency for certification	Continuous reactor + fixed blend recipe; CPCRI/KAU validation

20. Assumptions & disclaimer

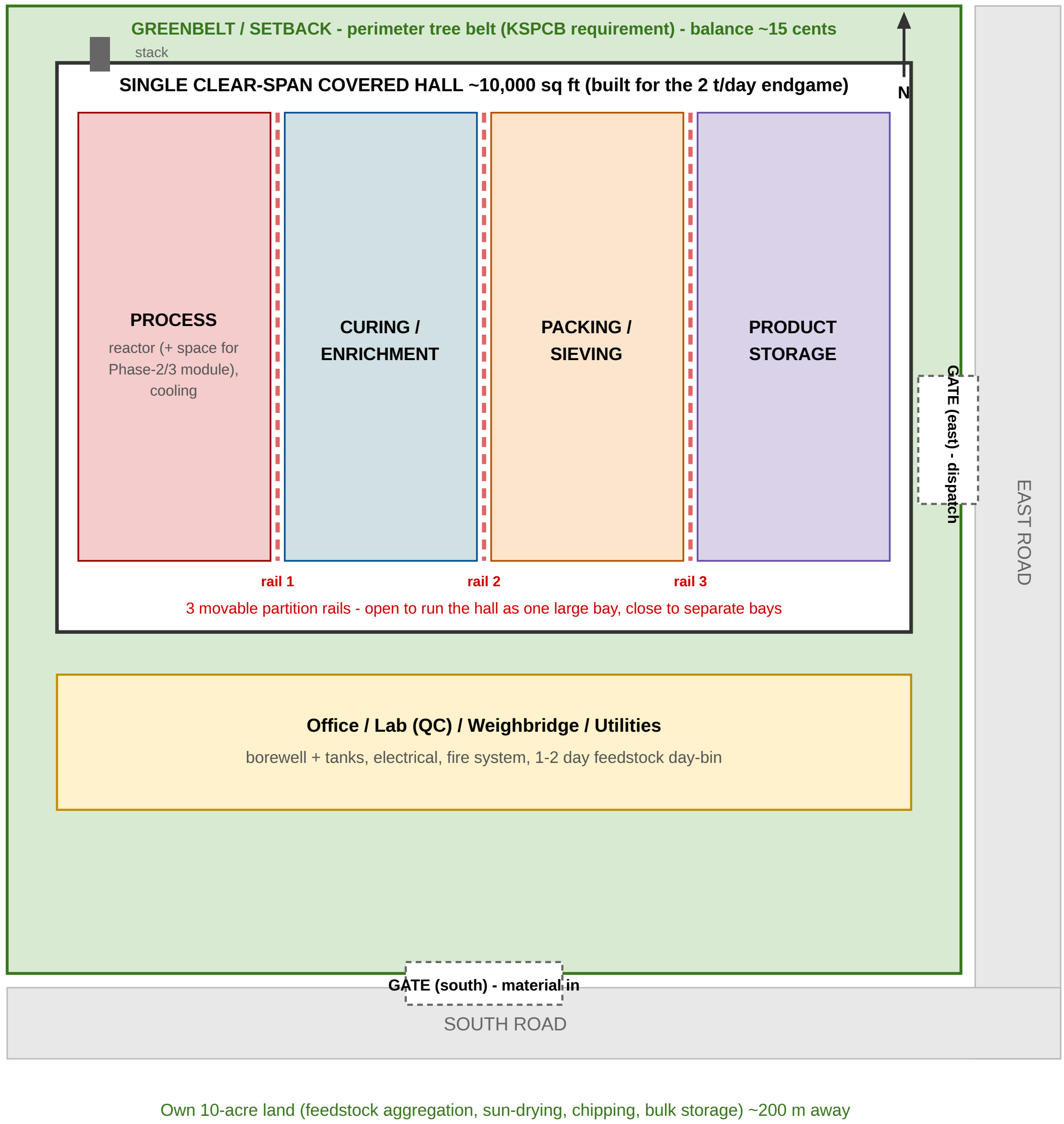
All costs, yields, prices and returns are **planning-grade indicative estimates** (mid-2026 market references) for scheme application and bank appraisal. They must be confirmed by: (a) a laboratory proximate analysis of the actual feedstock; (b) formal machinery and hall-construction quotations; (c) written confirmation of ESS/AIF eligibility and convergence from DIC Palakkad and the lending bank; (d) a **written KSPCB categorisation (Green) confirmation**; and (e) a carbon-credit LOI and export-certification (EBC / BICON) confirmation before export/carbon revenue is counted. Regulatory categories and subsidy rates are subject to the prevailing KSPCB, DIC and scheme guidelines at the time of application.

Annexure A — Plot & hall layout (38-cent corner plot; roads on south & east)

`plot-layout.svg` (open in any browser) shows the layout to scale. Text schematic below:

Plot & Hall Layout - 38-cent corner plot (roads on south & east)

Biochar & enriched bio-fertilizer unit - single clear-span ~10,000 sq ft hall with 3 movable partition rails (2 t/day-ready); schematic, not to scale

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+==================== 38-CENT CORNER PLOT ====================+
|  GREENBELT / SETBACK (perimeter tree belt)                  |
|   +-----------------------------------------------------+   |
|   |   SINGLE CLEAR-SPAN COVERED HALL  ~10,000 sq ft    |   |
|   |   +-----+ R +-----+ R +-----+ R +-----+            |   |
|   |   | bay | a | bay | a | bay | a | bay |            |   |
|   |   |  1  | i |  2  | i |  3  | i |  4  |            |   |
|   |   |proc.| l |cure | l |pack | l |store|            |   |
|   |   +-----+   +-----+   +-----+   +-----+            |   |
|   |   3 movable partition rails (R) - open = one        |   |
|   |   large bay; close = separate bays                  |   |
|   +-----------------------------------------------------+   |
|  [stack]   +------------------------------------+           |
|            | Office / Lab / Weighbridge / Util |  [GATE-E]==|== EAST ROAD
|            +------------------------------------+           |
+==============================[GATE-S]=====================+
                                   ||
                         ======= SOUTH ROAD =======
     Own 10-acre land (feedstock, sun-drying, storage) ~200 m -->
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Notes: the plot is a near-square **corner plot with public roads on the south and east**, allowing a **material-in gate (south)** and a **dispatch gate (east)**. The **~10,000 sq ft single clear-span covered hall** occupies ~23 of the 38 cents; **3 movable partition rails** (sliding partitions on floor tracks) let the operator **open the whole floor as one large bay, or close it into separate process, curing, packing and storage bays**, resized as production ramps. Office/lab/weighbridge and the emission stack sit at the frontage; the balance of the plot is greenbelt, parking and setbacks. All bulk feedstock handling, sun-drying and storage are on the **own 10-acre land ~200 m away**, so only a 1–2 day day-bin sits in the hall.

Annexure B — Term-loan amortisation (Phase-1)

Assumptions: term loan **₹74 L**; effective rate **6% p.a.** (9% − 3% AIF subvention); tenure **7 years**; **6-month construction moratorium** (interest served, no principal in Year 1); **Kerala ESS ~₹40 L received ~Month 12 and applied as principal prepayment**, leaving ~₹34 L to amortise over Years 2–7 (EAI ≈ **₹6.92 L/yr**, ≈ ₹0.58 L/month).

Year	Opening (₹L)	Interest @6% (₹L)	Principal repaid (₹L)	ESS prepayment (₹L)	Closing (₹L)
1	74.00	4.44	0.00 (moratorium)	40.00	34.00
2	34.00	2.04	4.88	–	29.12
3	29.12	1.75	5.17	–	23.95
4	23.95	1.44	5.48	–	18.47
5	18.47	1.11	5.81	–	12.66
6	12.66	0.76	6.16	–	6.50
7	6.50	0.39	6.53	–	~0.00
Total		**~11.9**	**~34.0**	**40.0**	

AIF benefit: the 3% subvention saves ~₹6 L of interest over the loan life; the loan is **collateral-free** under CGTMSE — the land is not mortgaged.

Annexure C — Month-by-month cash flow, Year 1 (Phase-1 ramp)

Assumptions (physical-sales base case): commissioning at **Month 7** (larger hall build); ramp 30% → 100% by Month 12. Full-capacity monthly revenue **₹4.38 L** (₹52.5 L/yr); operating cost ≈ **₹1.2 L fixed + ₹1.3 L variable at 100%**; interest served during moratorium ≈ **₹0.37 L/month** (on ₹74 L). Capex is funded by equity + term loan (not shown in operating flow). ESS ₹40 L inflow (~Month 12) is a financing item applied to loan prepayment (Annexure B).

Month	Phase	Capacity	Revenue (₹L)	Op. cost (₹L)	EBITDA (₹L)	Interest (₹L)	Net op. cash (₹L)
1–6	Construction / pre-op	–	0.00	trial only	–	0.37/mo	interest served from loan
7	Commissioning	30%	1.31	1.59	−0.28	0.37	−0.65
8	Ramp	45%	1.97	1.79	0.18	0.37	−0.19
9	Ramp	60%	2.63	1.98	0.65	0.37	0.28
10	Ramp	75%	3.28	2.18	1.10	0.37	0.73
11	Ramp	90%	3.94	2.38	1.56	0.37	1.19
12	Full	100%	4.38	2.50	1.88	0.37	1.51

Steady state (Year 2+, Phase-1): monthly revenue ₹4.38 L − opex ₹2.50 L = **EBITDA ₹1.88 L**; less EMI ₹0.58 L → **net cash ~₹1.30 L/month (~₹15.6 L/yr)** before tax and depreciation.

Debt-service coverage (Year 2): EBITDA ≈ ₹22.5 L ÷ debt service ≈ ₹6.92 L → **DSCR ≈ 3.3** (bankable).

Upside: adding realized carbon revenue (~₹33 L/yr at Phase-1) and ramping capacity inside the same hall lifts EBITDA sharply — kept out of the base case above pending a signed aggregator LOI.

Annexure D — Machinery quotation-comparison template (fillable, Phase-1 ~0.5 t/day)

How to use: send this to at least **two local vendors (Coimbatore/Pollachi hub)** and **one national continuous-technology vendor**. Ask each to quote a complete **~0.5 t/day plant** with **pre-provisioned tie-ins for Phase-2/3 modules**, state guaranteed emission values and yield, then carry the total into **Section 11 — Cost of project**.

Table D-1 — Scope & required specification (issue to vendors)

#	Equipment	Sized for	Required specification / notes	→ Cost item
1	Pyrolysis / carbonization reactor	**~0.5 t/day** (~100 kg/hr feed), expandable	Continuous/semi-continuous; multi-feedstock (husk, frond, shell, coir, sawdust); yield ≥25–30% dry basis; pre-laid tie-ins for Phase-2/3 module	#2
2	Feed system (hopper, screw/belt, metering)	0.5 t/day	Variable-speed; suited to shredded mixed biomass	#2
3	Waste-heat feed dryer	0.5 t/day	Uses pyrolysis/syngas waste heat; inlet moisture up to ~25%	#2
4	Emission control — afterburner/flare + cyclone + alkaline scrubber + mist eliminator + ID fan + stack	0.5 t/day (up-ratable)	Must meet **KSPCB Green / 02.12.2025 guideline** stack norms; vendor to state guaranteed emission values & stack height	#3

#	Equipment	Sized for	Required specification / notes	→ Cost item
5	Char cooling / quench + discharge	0.5 t/day	Sealed cooling; moisture/temperature controlled	#4
6	Sieving / screening	0.5 t/day	Grade separation (powder/granular)	#4
7	Enrichment mixer / blender	0.5 t/day	For char + manure/neem/rock-P + microbial slurry	#4
8	Weighing + bagging	0.5 t/day	25 kg bags + FIBC jumbo option	#4
9	Control panel / instrumentation	0.5 t/day	Temp/pressure monitoring, interlocks, e-stop	—
10	Erection, commissioning & operator training	—	Include site supervision & trial run	#2–#4
11	Freight, insurance, unloading	—	To site, Eruthenpathy	#2–#4
12	Taxes (GST)	—	State applicable rate separately	—

Table D-2 — Price comparison (fill from vendor quotes, ₹)

#	Equipment (as above)	Vendor A: ______	Vendor B: ______	Vendor C: ______
1	Pyrolysis reactor (0.5 t/day, expandable)	₹______	₹______	₹______
2	Feed system	₹______	₹______	₹______
3	Waste-heat dryer	₹______	₹______	₹______
4	Emission control train	₹______	₹______	₹______
5	Char cooling / quench	₹______	₹______	₹______
6	Sieving / screening	₹______	₹______	₹______
7	Enrichment mixer	₹______	₹______	₹______
8	Weighing + bagging	₹______	₹______	₹______
9	Control panel	₹______	₹______	₹______
10	Erection, commissioning, training	₹______	₹______	₹______
11	Freight, insurance	₹______	₹______	₹______
	Sub-total (ex-GST)	₹______	₹______	₹______
12	GST	₹______	₹______	₹______
	TOTAL (incl. GST)	**₹______**	**₹______**	**₹______**
	Target ≈ ₹22 L for reactor+dryer+emission (items #2–#3 of Section 11)			

Table D-3 — Commercial-terms comparison (fill from vendor quotes)

Parameter	Vendor A	Vendor B	Vendor C
Make / model offered			
Rated capacity (kg/hr feed; t/day biochar)			
Guaranteed biochar yield (% dry basis)			
Expandability to 1–2 t/day (tie-in provisions)			
Material of construction (reactor)			
Connected electrical load (kW)			
Guaranteed stack emissions vs KSPCB Green/guideline norms			
Feedstock types accepted (husk/frond/shell/coir/sawdust)			
Delivery period (weeks from PO)			
Warranty (months)			
After-sales / spares — distance from site			
Payment terms			
Training & commissioning included?			
References / plants in operation (contactable)			

Evaluation guidance: weight **emission-compliance (for Green CTO), guaranteed yield, feedstock flexibility, expandability in place, and after-sales proximity** (favouring the Coimbatore/Pollachi hub). A clean, low-emission continuous unit satisfies the Green/guideline norms and produces consistent, certifiable biochar for domestic sale and premium export.